Southeast University

Introduction to Software Engineering

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Outline

Chap 6. Human Aspects of Software Engineering

- 1. Traits of Successful Software Engineers
- 2. Behavioral Model for Software Engineering
- 3. Software Teams
  - Team Leader
  - Team Roles
  - Effectiveness vs. Toxicity（毒性）
  - Team Structure
  - Agile Teams
- 4. Team Coordination & Communication

Traits of Successful Software Engineers

- Sense of individual responsibility
- Acutely aware of the needs of team members and stakeholders
- Brutally honest about design flaws（缺陷） and offers constructive criticism
- Resilient under pressure
- Heightened sense of fairness
- Attention to detail
- Pragmatic（务实）

- 个人责任感
- 敏锐地意识到团队成员和利益相关者的需求
- 对设计缺陷（缺陷）直言不讳并提供建设性的批评
- 在压力下坚韧不拔
- 高度的公平感
- 注重细节
- 务实（务实）

Software Teams

How to lead?

How to collaborate?

How to organize?

How to motivate?

How to create good ideas?

Team Leader

- The MOI Model
  - **Motivation.** The ability to encourage (by "push or pull") technical people to produce to their best ability.
  - **Organization.** The ability to mold existing processes (or invent new ones) that will enable the initial concept to be translated into a final product.
  - **Ideas or innovation.** The ability to encourage people to create and feel creative even when they must work within bounds established for a particular software product or application.

边界跨越团队角色
Boundary Spanning Team Roles

- **Ambassador（外联员） – represents team to outside constituencies**
- **Scout（侦查员） – crosses team boundaries to collect information**
- **Guard （守护员）– protects access to team work products**
- **Sentry （安检员）– controls information sent by stakeholders**
- **Coordinator（协调员） – communicates across the team and organization**

有效的软件团队属性
Effective Software Team Attributes

- **Sense of purpose**目标意识
- **Sense of involvement**参与意识
- **Sense of trust**信任意识
- **Sense of improvement**进步意识
- **Diversity of team member skill sets**技能多样性

Avoid Team "Toxicity" (毒素）

狂热的工作氛围

- **A frenzied work atmosphere** in which team members waste energy and lose focus on the objectives of the work to be performed.

高挫折感（沮丧）

- **High frustration** （沮丧）caused by personal, business, or technological factors that cause friction among team members.

零散或协调不良的程序

- "**Fragmented or poorly coordinated procedures**" or a poorly defined or improperly chosen process model that becomes a roadblock to accomplishment.

角色定义不清

- **Unclear definition of roles** resulting in a lack of accountability and resultant finger-pointing.

持续和反复的失败暴露

- "**Continuous and repeated exposure to failure**" that leads to a loss of confidence and a lowering of morale.

组织范式
Organizational Paradigms

封闭范式
- **closed paradigm**—structures a team along a traditional hierarchy of authority

随机范式
- **random paradigm** —structures a team loosely and depends on individual initiative of the team members

开放范式
- **open paradigm** —attempts to structure a team in a manner that achieves some of the controls associated with the closed paradigm but also much of the innovation that occurs when using the random paradigm

同步范式
- **synchronous paradigm** —relies on the natural compartmentalization of a problem and organizes team members to work on pieces of the problem with little active communication among themselves

suggested by Constantine [Con93]

影响团队结构的因素
Factors Affecting Team Structure

The following factors must be considered when selecting a software project team structure ...

- the **difficulty of the problem** to be solved
- the **size of the resultant program(s)** in lines of code or function points
- the **time that the team will stay together** (team lifetime)
- the **degree to which the problem can be modularized**
- the **required quality and reliability** of the system to be built
- the **rigidity of the delivery date**
- the **degree of sociability** (communication) required for the project

- 要解决的问题的难度
- 所产生的程序的规模（以代码行数或功能点数计算）
- 团队在一起的时间（团队寿命）
- 问题模块化的程度
- 要构建系统的质量和可靠性要求
- 交付日期的刚性
- 项目所需的社交程度（沟通）

通用敏捷团队

Generic Agile Teams

- **Stress individual competency** coupled with **group collaboration** as critical success factors
- **People** trump process and **politics** can trump people
- Agile teams as self-organizing and have many structures
  - An **adaptive team structure**
  - Uses elements of Constantine's random, open, and synchronous structures
  - Significant autonomy（自主权）
- **Planning is kept to a minimum** and constrained only by business requirements and organizational standards

- 强调个人能力与团队协作相结合作为关键成功因素
- 人比过程更重要，而政治可能会凌驾于人之上
- 计划保持在最低限度，并仅受业务需求和组织标准的约束

- 敏捷团队是自组织的，并且有多种结构
  - 自适应团队结构
  - 使用Constantine的随机、开放和同步结构的元素
  - 具有显著的自主权（自主管理）

XP团队价值
XP Team Values

- **Communication** – close informal verbal communication among team members and stakeholders and establishing meaning for metaphors as part of continuous feedback
- **Simplicity** – design for immediate needs nor future needs
- **Feedback** – derives from the implemented software, the customer, and other team members
- **Courage** – the discipline to resist pressure to design for unspecified future requirements
- **Respect** – among team members and stakeholders

- **沟通** —— 团队成员和利益相关者之间的密切非正式口头交流，并在持续反馈中建立隐喻的意义
- **简洁** —— 为当前需求设计，而不是未来需求
- **反馈** —— 来源于实施软件、客户和其他团队成员
- **勇气** —— 抵制设计未明确需求的未来要求的压力的纪律
- **尊重** —— 团队成员和利益相关者之间

Scrum 框架

3种角色

- 产品负责人
- Scrum Master
- 开发团队

3种工件

- 产品待办列表 Product Backlog
- Sprint待办列表 Sprint Backlog
- 增量 Increment

5种事件

- Sprint
- Sprint计划会议 Sprint Plannin
- 每日Scrum站会 Daily Scrum
- Sprint评审会议 Sprint Review
- Sprint回顾会议 Sprint Retrosp

5种价值观

- 承诺 Commitment
- 勇气 Courage
- 专注 Focus
- 开放 Openness
- 尊重 Respect

Scrum 团队模型（三种角色）

Scrum Master

- 保证Scrum团队可以遵守；Scrum的价值，实践和规范
- 帮助Scrum团队和组织采用Scrum模式进行项目流程组织；
- 指导并带领团队变得更加高效，实现更高质量；
- 保护团队不要受到外界因素的干扰；
- 保证各个不同角色之间的良好写作，消除障碍；
- 帮助PO更好地利用团队的能力；
- 不要管理团队。

Product Owner

- PO 是一个人并只能由一个人来担任；
- 负责管理产品待办事项表（Product Backlog）并保证其对于客户和团队保持透明度；
- 对产品代办事项表进行优先级排序；
- 与团队一起来进行工作量估算；
- 对于项目的成功负责并保证投资回报率（ROI）。

团队

- 最佳团队大小：5-9 人；
- 多功能团队：程序员，测试人员，设计师，数据库管理员和架构师；
- 保证团队成员全职参与开发
- 自我管理，没有头衔之分，不组建子团队；
- 成员更替只能在迭代之间进行，最佳方式是在发布之间进行。

团队协调与沟通
Team Coordination & Communication

- *Formal, impersonal approaches* **include software engineering documents and work products (including source code), technical memos, project milestones, schedules, and project control tools, change requests and related documentation, error tracking reports, and repository data.**
- *Formal, interpersonal procedures* **focus on quality assurance activities applied to software engineering work products. These include status review meetings and design and code inspections.**

- **正式、非人际的方法** 包括软件工程文档和工作产品（包括源代码）、技术备忘录、项目里程碑、时间表和项目控制工具、变更请求和相关文档、错误跟踪报告和存储库数据。
- **正式、有人际的方法** 专注于应用于软件工程工作的质量保证活动。这些包括状态评审会议和设计及代码检查。

Team Coordination & Communication

- *Informal, interpersonal procedures* include group meetings for information dissemination and problem solving and “collocation of requirements and development staff.”
- *Electronic communication* encompasses electronic mail, electronic bulletin boards, and by extension, video-based conferencing systems.
- *Interpersonal networking* includes informal discussions with team members and those outside the project who may have experience or insight that can assist team members.

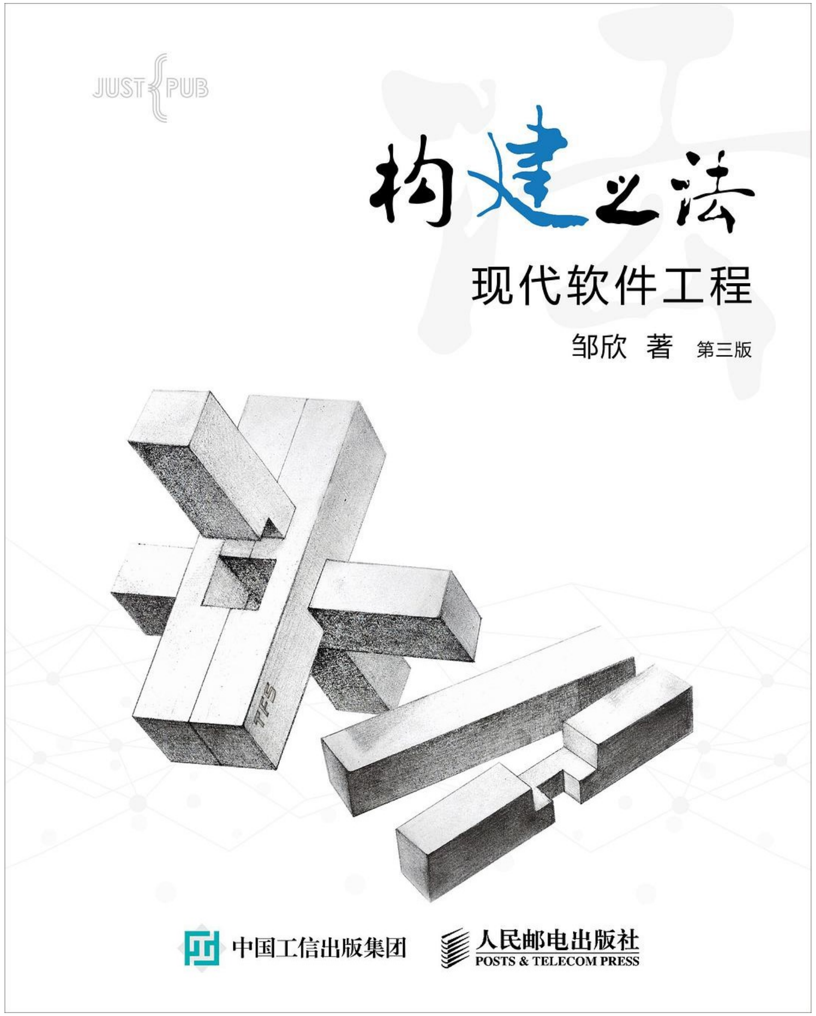
- **非正式、有人际的方法** 包括信息传播和问题解决的团队会议，以及“需求和开发人员的同地化”。
- **电子通信** 包括电子邮件、电子公告板，以及扩展的基于视频的会议系统。
- **人际网络** 包括与项目内外可能具有经验或见解的人进行非正式讨论，帮助团队成员。

社交媒体的影响

Impact of Social Media

- **Blogs** – can be used share information with team members and customers
- **Microblogs** (e.g. Twitter) – allow posting of real-time messages to individuals following the poster
- **Targeted on-line forums** – allow participants to post questions or opinions and collect answers
- **Social networking sites** (e.g. Facebook, LinkedIn) – allows connections among software developers for the purpose of sharing information
- **Social book marking** (e.g. Delicious, Stumble, CiteULike) – allow developers to keep track of and share web-based resources

- **博客** —— 可以用来与团队成员和客户共享信息
- **微博（如Twitter）** —— 允许向关注者发布实时消息
- **针对性在线论坛** —— 允许参与者发布问题或意见并收集答案
- **社交网络网站（如Facebook、LinkedIn）** —— 允许软件开发人员之间建立连接，以分享信息
- **社交书签（如Delicious、Stumble、CiteULike）** —— 允许开发人员跟踪和共享基于网络的资 源



- 邹欣现任微软Windows中国工程团队首席研发总监。1996—2003年，邹欣在微软Outlook团队从事开发工作，2003—2005年，他在微软内部质量工具团队和Visual Studio团队负责软件项目管理工具的开发。2005—2012年，他担任微软亚洲研究院技术创 新组研发主管，负责研究成果的产品化和创新项目。2012—2014年，他担任微软亚洲互联网工程院首席研发总监，负责必应搜索客户端、必应输入法、必应词典等产品。

使用云进行软件工程

Software Engineering using the Cloud

- Benefits
  - Provides access to all software engineering work products
  - Removes device dependencies and available every where
  - Provides avenues for distributing and testing software
  - Allows software engineering information developed by one member to be available to all team members
- Concerns
  - Dispersing cloud services outside the control of the software team may present reliability and security risks
  - Potential for interoperability problems becomes high with large number of services distributed on the cloud
  - Cloud services stress usability and performance which often conflicts with security, privacy, and reliability

- 好处
  - 提供访问所有软件工程工作产品的权限
  - 消除设备依赖性，随时随地可用
  - 提供分发和测试软件的途径
  - 允许一个成员开发的软件工程信息对所有团队成员可用
- 担忧
  - 在软件团队控制之外分散云服务，可能会带来可靠性和安全风险
  - 随着云上分布的服务数量增加，互操作性问题的潜在性增大
  - 云服务强调可用性和性能，这往往与安全性、隐私和可靠性冲突

协作工具
Collaboration Tools

- Collaborative Development Environments （协作开发环境）
  - Namespace that allows secure, private storage or work products
  - Calendar for coordinating project events
  - Templates that allow team members to create artifacts that have common look and feel
  - Metrics support to allow quantitative assessment of each team member’s contributions
  - Communication analysis to track messages and isolates patterns that may imply issues to resolve
  - Artifact clustering showing work product dependencies

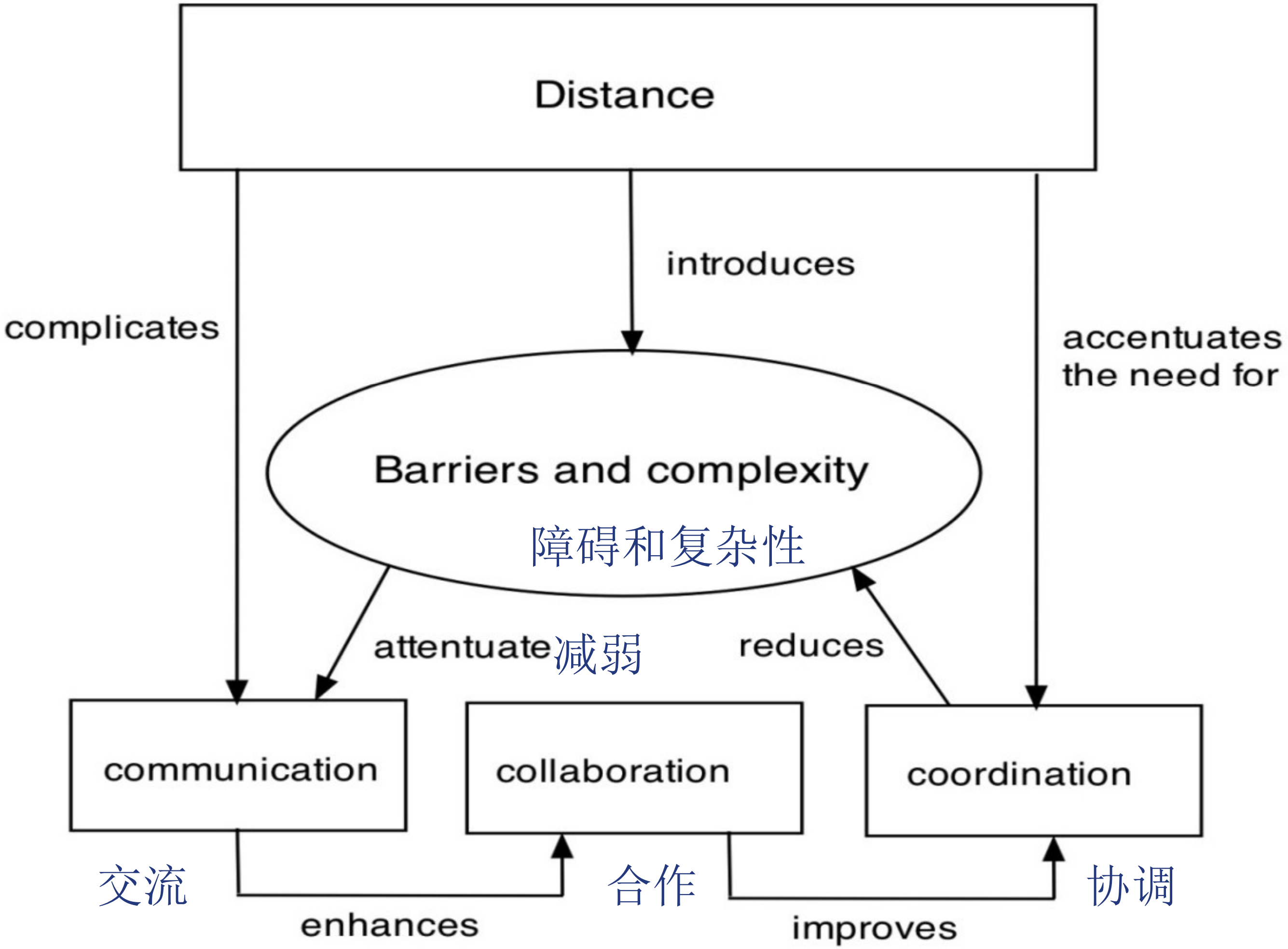
- 协作开发环境（协作开发环境）
  - 命名空间，允许安全、私密的存储或工作产品
  - 用于协调项目事件的日历
  - 允许团队成员创建具有共同外观和感觉的工件的模板
  - 支持允许对每个团队成员贡献进行定量评估的指标
  - 用于跟踪消息并隔离可能表示问题的模式的通信分析
  - 显示工作产品依赖性的工件聚类

团队决策的复杂性
Team Decisions Making Complications

- Problem complexity
- Uncertainty and risk associated with the decision
- Work associated with decision has unintended effect on another project object (law of unintended consequences)
- Different views of the problem lead to different conclusions about the way forward
- Global software teams face additional challenges associated with collaboration, coordination, and coordination difficulties

- 问题复杂性
- 与决策相关的不确定性和风险
- 与决策相关的工作对另一个项目对象产生意外影响（意外后果法则）
- 对问题的不同看法导致对前进方向的不同结论
- 全球软件团队面临的额外挑战包括协作、协调和沟通困难

Factors Affecting Global Software Development Team



Book Recomendation

- Peopleware
  - Tom DeMarco /Timothy Lister